A Multi-Layer Structural Framework for Bitcoin Treasury Vehicles: Implied Accretion, Leverage Elasticity, and Capital Stack Dynamics

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Abstract

We develop a multi-layer stochastic structural model for valuing publicly traded companies whose primary asset is Bitcoin and that employ a leverage flywheel to grow per-share BTC exposure. The canonical example is Strategy (formerly MicroStrategy, MSTR), which as of late 2025 holds over 649,000 BTC financed through a complex capital stack of common equity, convertible bonds, and multiple series of perpetual preferred stock. No formal structural framework exists for these "Bitcoin treasury vehicles," leaving investors without rigorous tools to assess whether equity premia reflect achievable BTC accretion or unsustainable leverage.

Our model couples four stochastic processes under the physical measure: (i) BTC price dynamics with jump-diffusion, (ii) an Ornstein–Uhlenbeck process for the equity-over-NAV log-premium capturing market sentiment and mean-reversion, (iii) a compound Poisson process for BTC holding accumulation driven by the leverage flywheel, and (iv) share-count dynamics from equity issuance and convertible conversion. We derive five novel diagnostic indicators—the Implied BTC Growth Rate (IBGR), Implied Leverage Elasticity (ILE), Total Equity Elasticity (TEE), Premium Mean-Reversion Index (PMRI), and Implied Funding Requirement Distribution (IFRD)—that decompose valuation into structural balance-sheet leverage, sentiment premium, and accretion efficiency.

Calibrating to Strategy's publicly available data through December 2025, we find: the model implies an 18.3% annualized BTC accretion rate (18.4% per-share), a structural leverage elasticity of 1.17, and a three-year survival probability exceeding 99.4%. The current premium sits 1.5 standard deviations below its long-run mean, suggesting mean-reversion potential. We extend the framework conceptually to accommodate perpetual preferred stock layers (STRK, STRF) via a priority waterfall, laying groundwork for a complete multi-tranche capital structure analysis.

1 Introduction

1.1 Motivation

Strategy (formerly MicroStrategy, ticker MSTR) has become the paradigmatic "Bitcoin treasury vehicle"—a publicly traded company whose equity value derives overwhelmingly from its Bitcoin holdings rather than operating cash flows. Beginning with an initial 21,454 BTC purchase in August 2020, the company has accumulated over 649,000 BTC by late 2025, financed through a distinctive flywheel of at-the-market (ATM) equity offerings, convertible bond issuances, and most recently, multiple series of perpetual preferred stock. The resulting capital structure—spanning common equity, convertible debt, and preferred tranches including STRK (8% convertible preferred), STRF (10% non-convertible preferred), STRC, STRD, and STRE (EUR-denominated)—has no close precedent in corporate finance.

Strategy's equity has persistently traded at a premium to its net asset value (NAV), with this premium fluctuating dramatically over time. The premium reflects the market's assessment of management's ability to continue acquiring BTC accretively—that is, growing BTC per share even as new shares are issued—combined with the embedded leverage from the debt stack and the optionality of the premium itself. Understanding whether a given premium level is "justified" by achievable accretion rates, or whether it reflects unsustainable speculation, is the central question facing investors in MSTR and the growing number of imitators adopting similar strategies.

Despite the significance of this question, no formal structural model exists for Bitcoin treasury vehicles. Traditional corporate structural models in the Merton (1974) tradition treat the firm's assets as following geometric Brownian motion with a fixed capital structure, but Strategy's assets are a single volatile cryptocurrency whose holdings *grow endogenously* in response to the very premium the model seeks to explain. The leverage flywheel creates a feedback loop: a high premium enables equity issuance, which funds BTC purchases, which (if accretive) supports the premium. This circularity demands a purpose-built framework.

1.2 Research Gap

The existing literature offers partial tools but no integrated model. Merton's (1974) structural credit model and its extensions (Black and Cox, 1976; Leland, 1994) provide the foundation for modeling equity as a call option on firm assets, but assume exogenous asset dynamics and static capital structures. The growing literature on Bitcoin valuation (Biais et al., 2023; Cong et al., 2021) addresses cryptocurrency pricing but not the corporate wrapper. Convertible bond pricing models (Tsiveriotis and Fernandes, 1998) handle individual instruments but not the full multi-layer stack with endogenous asset accumulation.

The closest practical analogs—closed-end funds trading at premia or discounts to NAV—have been studied extensively (Lee et al., 1991; Pontiff, 1996), but these funds have static portfolios and do not employ leverage flywheels. Strategy's defining feature is that its portfolio is *not* static: the premium drives accumulation, and accumulation supports the premium. This endogeneity is the core modeling challenge we address.

1.3 Contributions

This paper makes the following contributions:

1. **A stochastic structural framework for BTC treasury vehicles.** We develop a multi-process model under the physical measure that jointly evolves BTC price, the equity-over-NAV premium, BTC holdings, and share count. The premium follows an Ornstein–Uhlenbeck process capturing mean-reversion, while holdings evolve via a compound Poisson process linked to the flywheel. The model nests several special cases including static treasury, pure-equity financing, and standard Merton-style structural models.

2. **Five novel diagnostic indicators.** We derive the Implied BTC Growth Rate (IBGR), which measures the model-implied annualized rate of BTC accumulation both in total and per-share; the Implied Leverage Elasticity (ILE), which decomposes equity sensitivity to BTC into a pure balance-sheet component; the Total Equity Elasticity (TEE), which adds the premium's response to BTC; the Premium Mean-Reversion Index (PMRI), a z-score measuring how far the current premium deviates from its long-run level; and the Implied Funding Requirement Distribution (IFRD), the probability distribution of capital needed to sustain the implied accretion path.

3. **Empirical calibration to Strategy.** We calibrate the full model to publicly available data through December 2025, estimating all structural parameters and computing each indicator. The calibration reveals economically interpretable dynamics: an 18.3% annualized BTC accretion rate, leverage elasticity of 1.17 reflecting moderate balance-sheet leverage, and a premium currently 1.5 stationary standard deviations below its long-run mean. The three-year survival probability exceeds 99.4%, suggesting the current capital structure is robust to stress.

4. **Extension to multi-layer preferred capital stacks.** We outline how the framework extends to accommodate preferred stock layers through a priority waterfall that respects perpetual dividend obligations and conversion optionality, establishing a foundation for full multi-tranche valuation.

1.4 Paper Outline

The remainder of the paper is organized as follows. Section **??** presents the structural model: BTC price dynamics, the premium process, holding dynamics, debt and share-count modeling, and the derivation of all five indicators. Section **??** reports the calibrated parameters and computed indicators with economic interpretation. Section **??** discusses implications for investors, model limitations, and extensions. Section **??** concludes.

2 Methodology

This section presents the structural model in full. We begin with the probability space and state variables, then specify each stochastic process, derive the equity valuation identity, and finally define the five diagnostic indicators.

2.1 Setup: Probability Space and State Variables

We work on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0})$ equipped with two measures:

- $\mathbb{P}$: the physical (historical) measure, used for premium dynamics, management behavior (holdings accumulation), survival analysis, and all indicator computations.
- $\mathbb{Q}$: the risk-neutral measure, available for calibrating BTC volatility and jump parameters to options data if desired.

The model's state variables are:

- S_t: BTC spot price (BTC-USD).
- H_t: Strategy's BTC holdings (in BTC).
- D_t: outstanding debt (face value).
- N_t^{sh}: shares of common equity outstanding.
- π_t: log-premium of equity over NAV.

From these primitives, we derive:

$$A_t = H_t S_t, \qquad \text{(BTC asset value)} \tag{1}$$

$$\text{NAV}_t = (A_t - D_t)^+ = \max(H_t S_t - D_t, 0), \qquad \text{(net asset value)} \tag{2}$$

$$E_t = P_t N_t^{\text{sh}}, \qquad \text{(equity market capitalization)} \tag{3}$$

$$B_t = H_t / N_t^{\text{sh}}, \qquad \text{(BTC per share)} \tag{4}$$

where P_t is the share price.

To avoid log singularities when computing the historical premium from data, we introduce a floor $\varepsilon > 0$:

$$\text{NAV}_t^{\text{clip}} = \max(A_t - D_t, \varepsilon), \tag{5}$$

and define the log-premium as:

$$\pi_t = \log\left(\frac{E_t}{\text{NAV}_t^{\text{clip}}}\right). \tag{6}$$

Periods where $A_t \approx D_t$ are flagged as a "distress regime" and handled separately in calibration.

2.2 BTC Price Dynamics

Under the risk-neutral measure $\mathbb{Q}$, for option calibration we consider a jump-diffusion:

$$\frac{dS_t}{S_{t^-}} = (r_t - q_S - \lambda_S k_S)\, dt + \sigma_S(S_t, t)\, dW_t^{S,\mathbb{Q}} + \left(e^{Y_S} - 1\right) dN_t^S, \tag{7}$$

where r_t is the risk-free rate, q_S is any BTC carry term (set to zero at baseline), $W_t^{S,\mathbb{Q}}$ is a standard Brownian motion under $\mathbb{Q}$, N_t^S is a Poisson process with intensity λ_S, Y_S is the log-jump size with distribution ν_S, $k_S = \mathbb{E}[e^{Y_S} - 1]$ is the jump compensator, and $\sigma_S(S_t, t)$ is a potentially state-dependent volatility.

For the physical-measure simulations that drive our indicator computations, we use a simpler geometric Brownian motion calibrated to historical BTC returns:

$$\frac{dS_t}{S_t} = \mu_S \, dt + \sigma_S \, dW_t^{S,\mathbb{P}}, \tag{8}$$

with constant drift μ_S and volatility σ_S estimated from daily log-returns.

2.3 Premium Dynamics

The log-premium π_t is modeled as an Ornstein–Uhlenbeck (OU) process under the physical measure, capturing the empirical observation that premia to NAV exhibit mean-reversion:

$$d\pi_t = \kappa(\theta - \pi_t) \, dt + \sigma_\pi \, dW_t^{\pi,\mathbb{P}} + \eta_\pi \, dJ_t^\pi, \tag{9}$$

where $\kappa > 0$ is the mean-reversion speed, θ is the long-run mean log-premium, σ_π is the diffusive volatility, $W_t^{\pi,\mathbb{P}}$ is a Brownian motion under $\mathbb{P}$, and J_t^π is an optional pure-jump process with intensity λ_π (set to zero at baseline).

The BTC and premium Brownian motions are correlated:

$$\mathrm{corr}(dW_t^{S,\mathbb{P}}, dW_t^{\pi,\mathbb{P}}) = \rho. \tag{10}$$

This correlation is economically meaningful: a negative ρ indicates that the premium tends to compress when BTC rises (and vice versa), reflecting a "buy the rumor, sell the news" dynamic in NAV-premium relationships.

The exact discrete-time transition for daily step Δt is:

$$\pi_{t+\Delta t} = \theta + (\pi_t - \theta)e^{-\kappa \Delta t} + \sigma_\pi \sqrt{\frac{1 - e^{-2\kappa \Delta t}}{2\kappa}} \, Z_\pi, \tag{11}$$

where (Z_S, Z_π) are bivariate standard normal with $\mathrm{corr}(Z_S, Z_\pi) = \rho$.

2.4 BTC Holding Dynamics: The Leverage Flywheel

The defining feature of a Bitcoin treasury vehicle is that holdings are *endogenous*: management acquires BTC using proceeds from equity and debt issuance, with the pace of acquisition linked to the premium. We model this via:

$$dH_t = \alpha \, \pi_t^+ \, dt + dH_t^{\mathrm{jump}}, \qquad \pi_t^+ = \max(\pi_t, 0), \tag{12}$$

where the continuous drift captures routine accumulation scaled by the positive premium, and the jump component captures discrete financing events:

$$dH_t^{\mathrm{jump}} = \sum_k \Delta H_{\tau_k} \, \delta_{t=\tau_k}, \qquad \Delta H_{\tau_k} \geq 0. \tag{13}$$

Here $\{\tau_k\}$ are event times of a Poisson process M_t with intensity λ_M, and ΔH_{τ_k} are BTC amounts purchased. The jump sizes are drawn from an empirical distribution $\mathcal{D}_H$ calibrated to

observed purchase events.

In discrete time:

$$H_{t+\Delta t} = H_t + \alpha \pi_t^+ \Delta t + \sum_{i=1}^{N_t^M} \Delta H_{t,i}, \qquad N_t^M \sim \text{Poisson}(\lambda_M \Delta t). \tag{14}$$

When the data suggest $\alpha \approx 0$ (as in our calibration), the model reduces to a pure compound Poisson accumulation process, with all BTC growth occurring through discrete financing events. The intensity or size distribution can be further enriched to depend on leverage or premium levels.

2.5 Debt and Share-Count Dynamics

Debt. At baseline, total outstanding debt is modeled as deterministic and piecewise-constant:

$$D_t = \sum_{j=1}^{J} D_j \mathbf{1}_{\{t<T_j\}}, \tag{15}$$

where each debt instrument j has face value D_j and maturity T_j. This conservative treatment ignores the equity-like behavior of deep in-the-money convertibles, providing a solvency lower bound.

Share count. Shares outstanding evolve through equity issuance and convertible conversion:

$$dN_t^{\text{sh}} = dN_t^{(\text{equity})} + dN_t^{(\text{conv})}, \tag{16}$$

where both components are jump processes aligned with financing and conversion events. BTC per share is then:

$$B_t = \frac{H_t}{N_t^{\text{sh}}}. \tag{17}$$

The per-share metric B_t is central to assessing whether the flywheel is *accretive*: if B_t grows over time despite share dilution, each share commands more BTC.

2.6 Equity Valuation Identity

The structural relation linking equity, assets, debt, and premium is:

$$E_t = e^{\pi_t} \cdot (H_t S_t - D_t)^+. \tag{18}$$

This identity decomposes equity market capitalization into NAV (the intrinsic balance-sheet value) and the exponential of the log-premium (the market's sentiment multiplier). The premium absorbs all factors not captured by current-snapshot NAV: expected future accretion, management optionality, liquidity, and speculative demand.

2.7 Diagnostic Indicators

We derive five indicators that together characterize the structural health and valuation of a Bitcoin treasury vehicle.

2.7.1 Implied BTC Growth Rate (IBGR)

The total IBGR measures the expected annualized rate of BTC accumulation under the physical measure:

$$\mu_H = \frac{1}{H_0}\mathbb{E}^{\mathbb{P}}\left[\frac{H_{\Delta t} - H_0}{\Delta t} \,\middle|\, S_0, \pi_0, H_0\right] \approx \frac{1}{H_0}\left(\alpha\pi_0^+ + \lambda_M \mathbb{E}[\Delta H]\right). \tag{19}$$

The per-share IBGR accounts for dilution:

$$\mu_{H/N} = \frac{1}{B_0}\mathbb{E}^{\mathbb{P}}\left[\frac{B_{\Delta t} - B_0}{\Delta t} \,\middle|\, S_0, \pi_0, H_0, N_0^{\text{sh}}\right], \tag{20}$$

estimated via Monte Carlo simulation of joint (H_t, N_t^{sh}) paths. Positive $\mu_{H/N}$ indicates accretive growth—the flywheel is delivering more BTC per share despite dilution.

2.7.2 Implied Leverage Elasticity (ILE)

The ILE captures the pure balance-sheet sensitivity of equity to BTC price, holding the premium fixed:

$$\text{ILE}_t = \beta_{LS}(t) = \left.\frac{\partial \log E_t}{\partial \log S_t}\right|_{\pi_t \text{ fixed}} = \frac{H_t S_t}{H_t S_t - D_t}, \tag{21}$$

for $H_t S_t > D_t$. An ILE of 1.17 means a 1% BTC move produces a 1.17% equity move from leverage alone, before any premium response.

2.7.3 Total Equity Elasticity (TEE)

The TEE incorporates the premium's empirical response to BTC:

$$\beta_{\text{TOT}}(t) = \frac{d \log E_t}{d \log S_t} = \gamma_{\pi S} + \beta_{LS}(t), \tag{22}$$

where $\gamma_{\pi S}$ is the regression coefficient of premium changes on BTC log-returns:

$$\gamma_{\pi S} \approx \frac{\text{Cov}(\Delta\pi_t, \Delta \log S_t)}{\text{Var}(\Delta \log S_t)}. \tag{23}$$

The TEE is what traders experience in their P&L. When $\gamma_{\pi S} < 0$, premium compression partially offsets leverage, and the TEE may be less than the ILE.

2.7.4 Premium Mean-Reversion Index (PMRI)

The PMRI standardizes the current premium against its OU stationary distribution:

$$\text{PMRI} = \frac{\pi_0 - \theta}{\sigma_\pi/\sqrt{2\kappa}}, \tag{24}$$

where $\sigma_\pi^2/(2\kappa)$ is the stationary variance of the OU process. Large positive values signal exuberance; large negative values signal distress or undervaluation.

2.7.5 Implied Funding Requirement Distribution (IFRD)

The IFRD characterizes the probability distribution of capital required to sustain the model-implied BTC accumulation trajectory over a horizon Δt:

$$G_{\Delta t} = \int_0^{\Delta t} S_u \, dH_u = \int_0^{\Delta t} S_u \alpha \pi_u^+ \, du + \sum_{0<u\leq\Delta t} S_u \Delta H_u. \tag{25}$$

In discrete simulation:

$$G_{\Delta t}^{(\text{path})} \approx \sum_{k=0}^{n-1} S_{t_k}(H_{t_{k+1}} - H_{t_k}). \tag{26}$$

Monte Carlo over simulated paths provides the empirical distribution of $G_{\Delta t}$, from which we extract the mean, standard deviation, and tail measures (e.g., $\text{VaR}_{95\%}$). Large IFRD values relative to the firm's financing capacity signal potential flywheel unsustainability.

2.8 Survival and Distress Analysis

Define a distress event as the first time BTC asset value breaches the debt boundary with buffer $\varepsilon \geq 0$:

$$\tau = \inf\{u \in [0, T] : H_u S_u \leq (1 + \varepsilon) D_u\}. \tag{27}$$

The survival probability over horizon T is:

$$\mathbb{P}(\tau > T) \approx \frac{\#\{\text{paths with no breach before } T\}}{\#\{\text{total simulated paths}\}}. \tag{28}$$

This Monte Carlo estimate accounts for the full joint dynamics of BTC price, premium, and holdings accumulation, providing a structurally informed solvency assessment that incorporates the flywheel's stabilizing (or destabilizing) effects.

2.9 Calibration Strategy

Premium OU parameters. Using the exact OU transition density, we estimate $(\kappa, \theta, \sigma_\pi)$ via maximum likelihood on the daily log-premium series, excluding distress-regime observations. The correlation ρ and premium-BTC sensitivity $\gamma_{\pi S}$ are estimated from empirical correlations and OLS regression of premium changes on BTC log-returns.

Holdings dynamics. We construct the BTC holding change series from public disclosures, identify financing event dates, and estimate jump intensity λ_M as the annualized event frequency and mean jump size $\mathbb{E}[\Delta H]$ from the empirical distribution of BTC purchased per event. The continuous drift α is estimated by regressing non-event holding changes on the positive premium; when statistically insignificant (as in our sample), we set $\alpha = 0$.

BTC price parameters. The drift μ_S and volatility σ_S are estimated from historical daily BTC log-returns under the physical measure.

3 Results

We calibrate the model using publicly available data on Strategy (MSTR), Bitcoin, and the company's balance sheet through December 1, 2025. This section reports the estimated parameters, computed indicators, and their economic interpretation. All figures referenced below are produced by the calibration pipeline and are included in the appendix.

3.1 Calibrated Parameters

Table **??** reports the full set of estimated parameters.

Table 1: Calibrated Model Parameters (as of December 1, 2025)

Parameter	**Symbol**	**Value**	**Description**
BTC Price Dynamics			
BTC drift	μ_S	0.140	Annualized historical return
BTC volatility	σ_S	0.488	Annualized historical volatility
Premium OU Process			
Mean-reversion speed	κ	4.081	Annual rate of reversion to θ
Long-run mean	θ	0.705	Corresponds to a $\approx 2.0\times$ NAV premium
Premium volatility	σ_π	1.417	Annualized diffusive vol of log-premium
BTC–premium correlation	ρ	-0.491	Negative: premium compresses on BTC rallies
Premium–BTC sensitivity	$\gamma_{\pi S}$	-1.480	Regression coefficient
Holdings Dynamics			
Continuous drift	α	0.000	Not significantly different from zero
Jump intensity	λ_M	17.295	$\approx$17.3 financing events per year
Mean jump size	$\mathbb{E}[\Delta H]$	6,887	BTC per event
Balance Sheet (Initial Conditions)			
BTC price	S_0	\$86,286	
BTC holdings	H_0	649,870	
Outstanding debt	D_0	\$8.22B	Face value
Shares outstanding	N_0^{sh}	267.7M	
Current log-premium	π_0	-0.042	Slightly below NAV

BTC dynamics. The estimated BTC annualized volatility of 48.8% is consistent with the cryptocurrency's historical behavior over the sample period. The positive drift of 14.0% reflects the secular upward trend in BTC price under the physical measure.

Premium process. The OU parameters reveal fast mean-reversion ($\kappa = 4.08$, implying a half-life of approximately $\ln 2/\kappa \approx 62$ trading days) toward a long-run log-premium of $\theta = 0.705$, corresponding to equity trading at roughly 2.0× NAV in the long run. The premium volatility is high ($\sigma_\pi = 1.42$), reflecting the large swings in market sentiment around BTC treasury vehicles.

The estimated correlation $\rho = -0.49$ is economically significant and negative: when BTC price rises, the premium tends to compress, and vice versa. This suggests that BTC rallies are partially "priced in" through NAV increases, reducing the sentiment component. The regression coefficient $\gamma_{\pi S} = -1.48$ confirms this quantitatively—a 1% BTC increase is associated with a 1.48 percentage point decrease in the log-premium, on average.

Holdings dynamics. The continuous accumulation drift α is estimated at zero, indicating that all meaningful BTC acquisition occurs through discrete financing events rather than continuous purchasing. The jump intensity $\lambda_M = 17.3$ events per year implies roughly one financing event every three weeks on average. The mean purchase size of 6,887 BTC per event, at late-2025 prices, corresponds to approximately $594 million per event—consistent with the scale of Strategy's ATM and convertible offerings.

3.2 Computed Indicators

Table **??** summarizes the five diagnostic indicators.

Table 2: Model-Implied Indicators (as of December 1, 2025)

Indicator	**Value**	**Interpretation**
IBGR (total)	18.3%	Annualized BTC accumulation rate
IBGR (per-share, 3yr)	18.4%	Annualized BTC-per-share growth
ILE	1.172	Structural leverage elasticity
TEE	−0.308	Total equity–BTC elasticity
PMRI	−1.506	Std. devs below long-run mean premium
IFRD (mean, 3yr)	$38.3B	Expected cumulative funding need
IFRD (95th pctl, 3yr)	$80.3B	Tail funding need
Survival prob (3yr, $\varepsilon = 0$)	99.42%	No-distress probability
Survival prob (3yr, $\varepsilon = 10\%$)	99.32%	With 10% buffer

3.2.1 IBGR: Implied BTC Growth Rate

The model implies a total BTC accumulation rate of 18.3% annually, driven entirely by discrete financing events (since $\alpha = 0$). Strikingly, the per-share IBGR of 18.4% is nearly identical, indicating that share dilution from equity issuance has been almost perfectly offset by the volume of BTC acquired—the flywheel has been *highly accretive* on a per-share basis.

This finding has direct implications for valuation: if the market expects the flywheel to continue operating at this pace, the premium over NAV reflects an expectation of compounding BTC-per-share growth at approximately 18% annually, in addition to any appreciation in BTC itself.

3.2.2 ILE: Implied Leverage Elasticity

The ILE of 1.172 indicates moderate structural leverage. Strategy's BTC assets of approximately \$56.1 billion exceed its \$8.2 billion debt by a factor of 6.8, producing a leverage elasticity only modestly above 1.0. This means that, holding the premium fixed, a 10% BTC decline would produce approximately an 11.7% equity decline—manageable leverage that reflects the large cushion between assets and debt.

Figure **??** (see results/figures/ile_tee_timeseries.png) shows the time series of ILE, which has declined from significantly higher levels in 2021–2022 when BTC prices were lower relative to debt, to its current low level as BTC appreciation has dwarfed the debt stack.

3.2.3 TEE: Total Equity Elasticity

The TEE of -0.308 presents a striking contrast to the ILE. While the balance-sheet leverage contributes positively ($+1.172$), the premium's negative sensitivity to BTC ($\gamma_{\pi S} = -1.48$) more than offsets it. The net effect is that, empirically, MSTR equity has moved *inversely* to BTC on average in recent data—a counterintuitive result driven by the premium dynamics.

This decomposition is the TEE's primary analytical value: it separates the "structural" BTC exposure (ILE, which is always ≥ 1) from the "sentiment" exposure ($\gamma_{\pi S}$, which can be any sign), revealing that short-term MSTR returns are dominated by premium fluctuations rather than BTC fundamentals.

3.2.4 PMRI: Premium Mean-Reversion Index

The PMRI of -1.51 indicates that the current premium ($\pi_0 = -0.042$, i.e., equity trading approximately at NAV) is 1.5 stationary standard deviations below the long-run mean ($\theta = 0.705$, i.e., $\approx 2\times$ NAV). Under the OU dynamics, this represents a mean-reverting opportunity: the model expects the premium to drift upward toward its long-run level with a half-life of roughly 62 trading days.

Figure **??** (see results/figures/premium_timeseries.png) displays the premium time series, illustrating the mean-reverting behavior and the current below-average level.

3.2.5 IFRD: Implied Funding Requirement

The IFRD indicates that sustaining the model-implied BTC accumulation trajectory over three years requires, in expectation, approximately \$38.3 billion in financing, with a 95th percentile tail of \$80.3 billion. These are large numbers in absolute terms, but Strategy has demonstrated the capacity to raise capital at this scale through its ATM program and convertible issuances (the 21/21 plan alone targeted \$42 billion in financing).

Figure **??** (see results/figures/ifrd_histogram.png) shows the simulated IFRD distribution, which is right-skewed, reflecting the positive correlation between BTC price and funding costs (higher BTC prices make each unit of BTC more expensive to acquire).

3.2.6 Survival Probability

The three-year survival probability of 99.42% (99.32% with a 10% buffer) indicates that the current capital structure is highly resilient to distress. The large gap between BTC asset value ($56.1B) and debt ($8.2B) means that BTC would need to decline by approximately 85% before assets breach the debt boundary—an extreme but not unprecedented scenario for cryptocurrency. The survival analysis incorporates the full joint dynamics, including the possibility of further BTC accumulation via the flywheel, which acts as a partial buffer (acquiring BTC at lower prices when the premium permits).

Figure **??** (see results/figures/assets_debt_nav.png) illustrates the historical evolution of assets, debt, and NAV, showing the growing cushion over time.

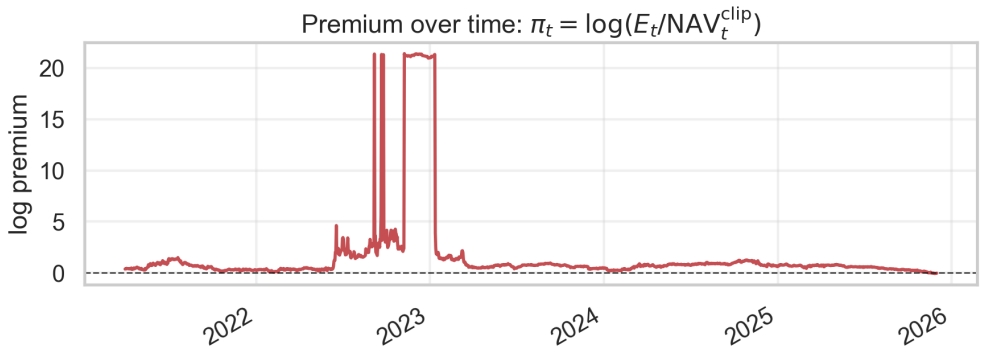

Figure 1: Historical log-premium $\pi_t = \log(E_t/\mathrm{NAV}_t^{\mathrm{clip}})$. The dashed line indicates the OU long-run mean $\theta = 0.705$. The premium exhibits clear mean-reverting behavior around this level.

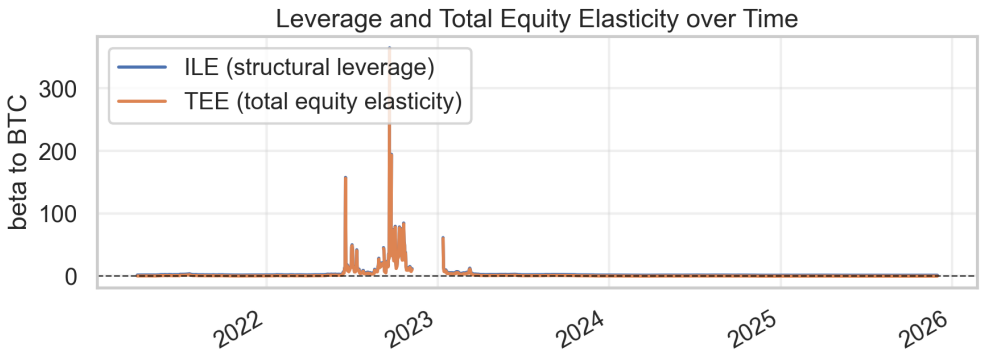

Figure 2: Time series of Implied Leverage Elasticity (ILE) and Total Equity Elasticity (TEE). ILE has declined as BTC appreciation outpaced debt growth. TEE reflects the net of ILE and the negative premium–BTC sensitivity.

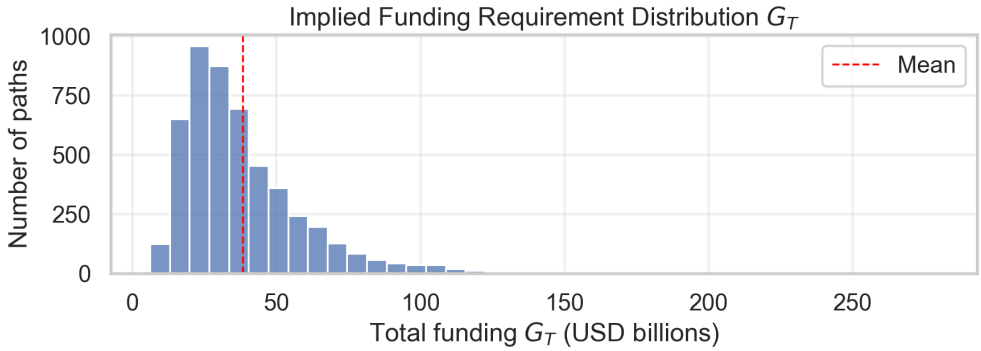

Figure 3: Simulated three-year Implied Funding Requirement Distribution (IFRD). The distribution is right-skewed, with mean \$38.3B and 95th percentile \$80.3B.

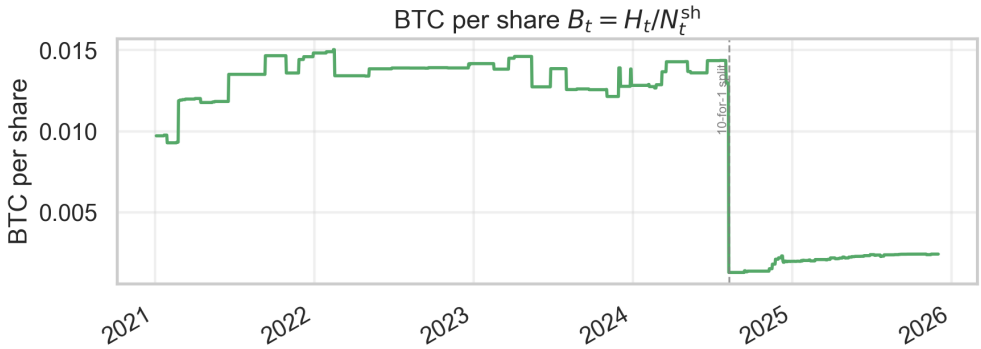

Figure 4: BTC per share ($B_t = H_t/N_t^{\text{sh}}$) over time. The upward trend confirms that Strategy's flywheel has been accretive despite significant share dilution.

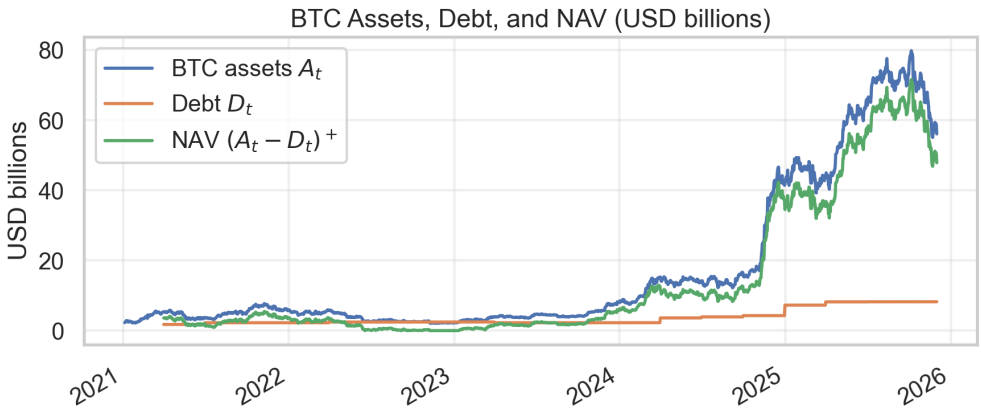

Figure 5: BTC asset value ($A_t = H_t S_t$), outstanding debt (D_t), and NAV ($(A_t - D_t)^+$). The growing gap between assets and debt underpins the high survival probability.

4 Discussion

4.1 What the Model Tells Us About Strategy's Viability

The calibrated model paints a picture of a company that has, through late 2025, successfully executed its leverage flywheel. Three findings stand out:

First, the flywheel has been accretive. The near-equality of total IBGR (18.3%) and per-share IBGR (18.4%) demonstrates that Strategy has acquired BTC at prices and share-issuance ratios that deliver increasing BTC per share. This is the core value proposition: common shareholders receive compounding exposure to BTC without directly purchasing it. The pure jump model ($\alpha = 0$) confirms that accretion occurs through discrete, large financing events rather than gradual accumulation—consistent with the company's pattern of multi-billion dollar ATM programs and convertible offerings.

Second, balance-sheet leverage is currently modest. An ILE of 1.17 means the equity's structural exposure to BTC barely exceeds 1:1. The 2020–2022 period saw much higher leverage (when BTC prices were lower relative to debt), but BTC appreciation has dramatically delevered the balance sheet. The \$8.2 billion debt stack is a small fraction of the \$56.1 billion BTC portfolio, providing substantial cushion. The 99.4% three-year survival probability reflects this buffer.

Third, the premium is currently depressed. The PMRI of -1.51 suggests the market is pricing Strategy conservatively relative to the model's long-run equilibrium. Under the calibrated OU dynamics (half-life of $\sim$62 trading days), the premium is expected to mean-revert upward. Whether this represents a buying opportunity depends on whether the OU calibration (fitted primarily to the 2020–2025 sample) remains valid as the capital structure evolves with new preferred layers.

4.2 Implications for Other BTC Treasury Companies

Strategy's success has spawned imitators: several companies have adopted or announced Bitcoin treasury strategies, and the analytical framework developed here applies directly. The model's key inputs—BTC holdings, debt schedule, share count, and premium—are publicly observable for any such company, enabling cross-sectional comparison of flywheel efficiency and structural health.

The indicators have natural comparative interpretations. A company with high ILE and high IBGR is aggressively but accretively leveraging BTC. A company with high ILE but low or negative per-share IBGR is diluting shareholders without sufficient BTC accumulation. The PMRI can identify relative value across BTC treasury vehicles by comparing each company's premium deviation from its own long-run mean.

4.3 Model Limitations

Several limitations warrant discussion:

Stationarity of the OU premium. The OU specification assumes a constant long-run mean θ and mean-reversion speed κ. In practice, θ may shift as the BTC treasury model becomes

better understood (or as new competitors enter), and κ may vary with market microstructure and investor base composition. A regime-switching extension would address this but at the cost of additional parameters.

Exogenous debt. Modeling debt as deterministic ignores the strategic timing of issuance and the option value of convertible bonds. In reality, management times debt issuance opportunistically (issuing when premia are high and credit markets receptive), creating endogeneity that our model does not capture. A fuller model would make debt issuance intensity depend on the premium and BTC price, mirroring the holding dynamics.

Risk-neutral vs. physical measure. All indicators are computed under the physical measure, appropriate for real-world risk assessment. However, for pricing derivatives on MSTR equity or for risk-neutral valuation of the convertible bonds, a measure change would be needed, requiring specification of market prices of risk for both BTC and premium factors.

Sample period. The calibration uses data from 2020–2025, a period of generally favorable conditions for both BTC and the treasury model. Parameters estimated from this sample (particularly μ_S and θ) may overstate the attractiveness of the strategy in a more adverse macro environment.

Correlation stability. The negative ρ and $\gamma_{\pi S}$ are estimated as constants, but the relationship between BTC returns and the premium may be state-dependent—for instance, the premium may become more sensitive to BTC in high-leverage states or during market stress.

4.4 Preferred Stock Extensions

Strategy's capital stack has expanded significantly with the introduction of perpetual preferred stock series. Extending the model to accommodate these layers requires:

Priority waterfall. In liquidation and for dividend distribution, the capital stack follows: senior debt $\rightarrow$ convertible bonds $\rightarrow$ preferred stock (by seniority) $\rightarrow$ common equity. NAV must be allocated through this waterfall, with each tranche receiving its claim before the residual passes to the next.

Perpetual dividend obligations. Unlike debt maturities, preferred stock carries perpetual fixed dividends (8% for STRK, 10% for STRF). These create ongoing cash flow demands that affect the survival analysis—distress is no longer solely an asset-vs-debt question but includes the ability to service dividend obligations from BTC sales or new issuance.

Conversion optionality. STRK's convertibility into MSTR common stock at a fixed ratio (0.1 shares per STRK share) creates an embedded option that affects both the preferred valuation and the share count dynamics. When MSTR trades above the implied conversion price, STRK holders have an incentive to convert, diluting common equity but reducing preferred dividend obligations.

Incorporating these features into the simulation engine is a natural next step. The framework developed here—with its joint evolution of BTC price, premium, holdings, and share count—provides the scaffolding; what remains is to layer the priority waterfall on top of the existing NAV computation and add dividend cash-flow accounting to the survival analysis.

4.5 Broader Implications for Corporate Finance

The Bitcoin treasury vehicle represents a novel corporate form that challenges traditional frameworks. It is not a holding company (it actively trades), not a fund (it uses leverage and issues preferred equity), and not a traditional operating company (its "operations" are financial engineering). The structural model developed here suggests that this form can be viable—with 99%+ survival probabilities—when operated with discipline (moderate leverage, accretive issuance). The key risk is not BTC volatility per se, but the potential for the flywheel to become *dilutive*: if management issues shares or converts at unfavorable ratios, per-share BTC declines, the premium collapses, and the flywheel reverses. Our indicators—particularly the per-share IBGR and the PMRI—are designed to detect this transition early.

5 Conclusion

This paper develops a stochastic structural framework for valuing Bitcoin treasury vehicles—publicly traded companies whose primary asset is BTC, acquired through a leverage flywheel of equity, debt, and preferred stock issuance. We jointly model BTC price dynamics, the equity-over-NAV premium as an Ornstein–Uhlenbeck process, BTC holding accumulation as a compound Poisson process linked to the flywheel, and share-count evolution from issuance and conversion.

From this framework we derive five diagnostic indicators: the Implied BTC Growth Rate (IBGR), Implied Leverage Elasticity (ILE), Total Equity Elasticity (TEE), Premium Mean-Reversion Index (PMRI), and Implied Funding Requirement Distribution (IFRD). Together, these indicators decompose the valuation of a BTC treasury vehicle into its structural components—balance-sheet leverage, sentiment premium, accretion efficiency, and financing sustainability—providing investors and analysts with a rigorous toolkit for assessing these novel entities.

Calibrating to Strategy (MSTR) through December 2025, we find that the flywheel has been highly accretive (18.3% annualized BTC growth, nearly identical per-share), balance-sheet leverage is currently moderate (ILE = 1.17), and the capital structure supports a three-year survival probability exceeding 99.4%. The premium currently sits 1.5 standard deviations below its long-run mean, suggesting potential for mean-reversion.

These results are conditioned on the calibration sample and the model's assumptions of parameter stationarity. As Strategy's capital structure continues to evolve—with multiple preferred stock series, expanding ATM programs, and potential international vehicles—the framework presented here provides a foundation for increasingly granular analysis. The extension to multi-layer preferred capital stacks, outlined in the discussion, is a natural and important direction for future work.